Tree exercises

1. Prove that a finite connected graph is a tree iff it has n−1 edges and no cycle.
2. A full binary tree has 31 leaves. Find its internal-node and edge counts and the minimum attainable height with root depth zero.
3. Use comparison decision trees to prove that sorting six distinct keys needs at least ten comparisons in the worst case. Evaluate $\lceil \log_2 6! \rceil$. Merge-sort [5,1,4,2,6,3], splitting odd lengths with a left half of floor(n/2), and count comparisons by level. If this input uses ten, does that prove optimal worst-case comparison count on all six-key inputs?
4. Build a Huffman tree for frequencies 2,3,5,7,11. Give a prefix code, weighted cost, and mean length.
5. Prove uniqueness of simple paths in a tree using the cycle formed by two different paths.
6. Draw a connected five-vertex graph with a one-child DFS root and a nonroot cut vertex. Apply the low-link cut criterion.
7. On edges AB, BC, CA, CD, DE, EF, FD, DG, start alphabetic DFS at A. Trace disc, low, parent, edge-ID pushes, and every complete pop. Find cuts, bridges, and vertex blocks. Then root at D; explain root-child and cut-criterion changes and outputs that must remain invariant.
8. Prove prefix-code lengths (1,2,2,3) impossible, and construct words for (2,2,3,3,3,3). For one symbol, distinguish optimal length zero from a specification requiring nonempty words.
9. A full ternary tree has thirteen internal nodes. Find total nodes, leaves, edges, and minimum height with root depth zero. Prove attainability rather than only a lower bound.
10. For positive weights $x \le y \le z$, prove the minimum binary-prefix weighted cost and give optimal words. Explain why z at depth two cannot improve the result. Evaluate merge cost and mean length for 2,5,9.
11. Prove every tree edge is a bridge and $\sum_v (\deg(v) - 2) = -2$. Find leaves when three vertices have degree three, one degree five, and every other nonleaf degree two. Explain the singleton exception to degree-one leaf counting.
12. A loop-free multigraph has vertices u, v and parallel IDs e_1,e_2. DFS discovers v from u along e_1. Use complete low-link rules to decide whether e_1 is a bridge and explain the error from skipping every edge to the parent vertex.

Tree exercises (continued)

13. A graph consists of the two triangles ABC and CDE sharing only C. List its bridges, articulation vertices and vertex blocks under the chapter convention, and draw the block-cut tree. Explain why deleting every bridge does not recover these vertex blocks. Then add one edge BD: prove what changes, using connectivity after vertex deletion rather than relying on the picture.
14. Four symbols have positive frequencies a:2, b:5, c:9, d:12. Initially use a:`0`, b:`10`, c:`110`, d:`111`.

    (a) Keep the shape, swap a,c, then swap b,d. List words, depths, W and its change at each step. Check whether the minimum-weight symbols are siblings.

    (b) Contract a,b to z:7. List reduced words and W, verifying the identity term by term. Is this reduced tree optimal? Solve the reduced problem and expand z to give optimal original words, W and mean length.

    (c) Compare W and height against fixed two-bit coding. Explain numerically why sibling minima alone and minimum height alone are insufficient certificates. Use the exchange and contraction lemmas to establish global optimality, not merely improvement over the initial tree.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study